

Case study 8:-

Mobile Recharge Billing:-Telecom operator wants to generate recharge invoices. Accept Recharge Amount and calculate GST (18%) and Final Amount Payable.

Algorithm – Mobile Recharge Billing

1. Start.
2. Declare variables: recharge, gst, total.
3. Input the Recharge Amount.
4. Calculate GST (18%):
 - $gst = recharge \times 18 / 100$
5. Calculate Total Amount:
 - $total = recharge + gst$
6. Display GST and Total Amount.
7. Stop.

Pseudocode:-

BEGIN

DECLARE recharge, gst, total AS FLOAT

INPUT recharge

$gst = recharge * 18 / 100$

$total = recharge + gst$

PRINT "GST = ", gst

PRINT "Total Amount = ", total

END

Test cases:- input/output combination.

```
PS C:\Users\khush\Desktop\PROJECT 1> gcc CASE_study_8.c -o CASE_study_8 ; if ($?) { .\CASE_study_8 }  
enter the amount of recharge 499  
total_amount=588.00final_GST = 89.00  
PS C:\Users\khush\Desktop\PROJECT 1> █
```

```
PS C:\Users\khush\Desktop\PROJECT 1> gcc CASE_study_8.c -o CASE_study_8 ; if ($?) { .\CASE_study_8 }  
enter the amount of recharge 649  
total_amount=765.00final_GST = 116.00  
PS C:\Users\khush\Desktop\PROJECT 1> █
```

```
PS C:\Users\khush\Desktop\PROJECT 1> gcc CASE_study_8.c -o CASE_study_8 ; if ($?) { .\CASE_study_8 }  
enter the amount of recharge 2200  
total_amount=2596.00final_GST = 396.00  
PS C:\Users\khush\Desktop\PROJECT 1> █
```